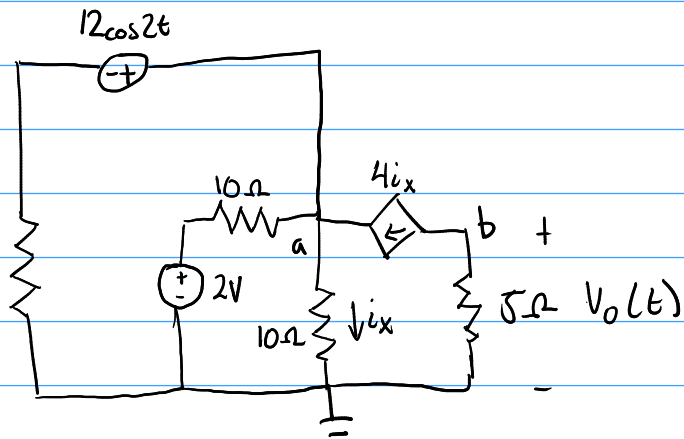


ECE 2040



$$V_A - V_B = 4i_x$$

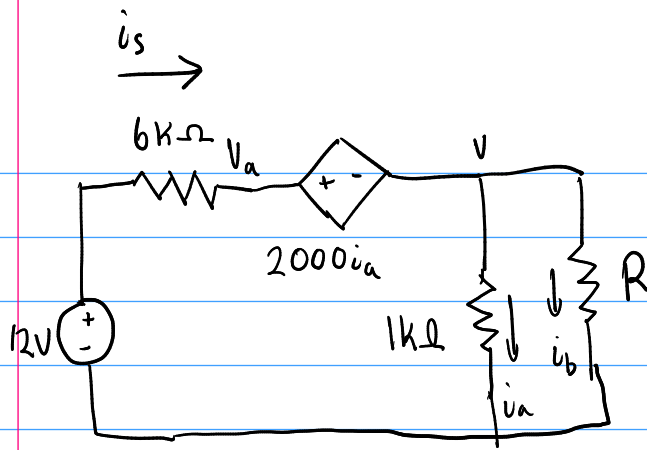
$$= 4 \frac{V_A}{10} = 0.4 V_A \quad V_B = 0.6 V_A$$

$$V_B = \frac{3}{85} V_1 + \frac{3}{17} V_2$$

$$i_x = \frac{V_A}{10}$$

$$V_o(t) = V_B(t) = \frac{36}{85} \cos(2t) + \frac{6}{17}$$

$$\approx 0.4235 \cos 2t + 0.3529$$



$$i_s = \frac{12 - 3}{6k\Omega}$$

$$i_s = i_a + i_b$$

$$i_a = \frac{v}{1k\Omega}$$

$$i_b = 2mA$$

$$2000i_a = 2V \quad 2mA = \frac{4}{3R + 2000}$$

$$v_a - 2V = V$$

$$v_a = 3V$$

$$R = 0\Omega$$